

Dmitrii Vasilenko

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RESEARCH INTERESTS

Theoretical Machine Learning, Bayesian Deep Learning, Neural Network Dynamics.

EDUCATION

Moscow Institute of Physics and Technology (MIPT) <i>M.Sc. in Computer Science (GPA: 4.9/5.0)</i>	Moscow, Russia <i>Sep. 2025 – Present</i>
Yandex School of Data Analysis (YSDA) <i>M.Sc. equivalent in Data Science (GPA: 5.0/5.0)</i>	Moscow, Russia <i>Sep. 2025 – Present</i>
Moscow Institute of Physics and Technology (MIPT) <i>B.Sc. in Applied Mathematics and Physics (GPA: 4.5/5.0)</i>	Moscow, Russia <i>Sep. 2021 – July 2025</i>

PUBLICATIONS

Jacobian Analysis of a Recurrent Transformer Block <i>AINL 2026</i>	Apr. 2026
– D. Vasilenko, I. Stepanov, V. Kasiuk, G. Karpeev, A. Grabovoy. Presented as a contributed talk at AINL 2026. – Derived explicit upper bounds on the spectral norm of the full-step Jacobian for Recurrent Transformers. – Theoretically proved and empirically validated that RMSNorm acts as a multiplicative damping factor, preventing gradient explosion and maintaining an $\mathcal{O}(1)$ Jacobian norm.	
Bensemble: A Modular Python Library for Bayesian Deep Learning <i>JOSS, in review</i> GitHub	Jul. 2026
– D. Vasilenko, M. Nabiev, F. Sobolevsky, V. Kasyuk, O. Bakhteev. – Open-source PyTorch library providing a unified interface for 8 Bayesian deep learning and neural network ensembling methods. – Implements Deep Ensembles, MC Dropout, Variational Inference, Laplace Approximation, Probabilistic Backpropagation, and Neural Ensemble Search. – Validated cross-method correctness on CIFAR-10 vs. SVHN out-of-distribution detection benchmarks.	
Sink vs. Diagonal Attention: Sharpened Cost Bounds and Comparison Regimes	Aug. 2026
– D. Vasilenko, A. Grabovoy. Under review at the AXIOM Workshop (Foundations of Efficient Deep Learning) @ NeurIPS 2026. – Proved an $\Omega(T)$ lower bound on the MLP cost of sink attention, conjectured but left open in prior work. – Characterized the sink cost as $\Theta(\kappa\sqrt{T} + T)$ at $d = T + 4$ and sharpened the diagonal lower bound by $\sqrt{2}$ via a variance identity.	

RESEARCH EXPERIENCE

Yandex R&D, YandexGPT Pretraining Team <i>Research Intern</i>	Moscow, Russia <i>Aug. 2026 – Present</i>
– Conducting research on LLM pretraining within the Knowledge Research group. – Investigating reasoning capabilities and knowledge acquisition in large language models.	
Intelligent Systems Department, MIPT <i>Graduate Researcher</i>	Moscow, Russia <i>Sep. 2025 – Present</i>
– Researching theoretical stability and dynamics of Recurrent Transformers under the supervision of Dr. Andrey Grabovoy. – Investigating Jacobian dynamics and spectral properties of deep Transformer components (MSA, FFN, RMSNorm).	
Moscow Institute of Physics and Technology (MIPT) <i>Undergraduate Researcher</i>	Moscow, Russia <i>Apr. 2024 – July 2025</i>
– Developed and implemented numerical solvers in C++ for hyperbolic partial differential equations as part of the Bachelor's thesis. – Analyzed systems of nonlinear equations using the Method of Characteristics.	

TECHNICAL SKILLS

Programming: Python, C++
ML Frameworks: PyTorch, NumPy
Developer Tools: Git, Docker, Linux, uv, LaTeX, GitHub Actions, pytest, Ruff
Natural Languages: Russian (Native), English (Advanced / C1), German (A2)